

→ active low decoder.

→ implement "fnc"

→ " of high order

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MUX & demux using low order MUX, demux

★ Decoder

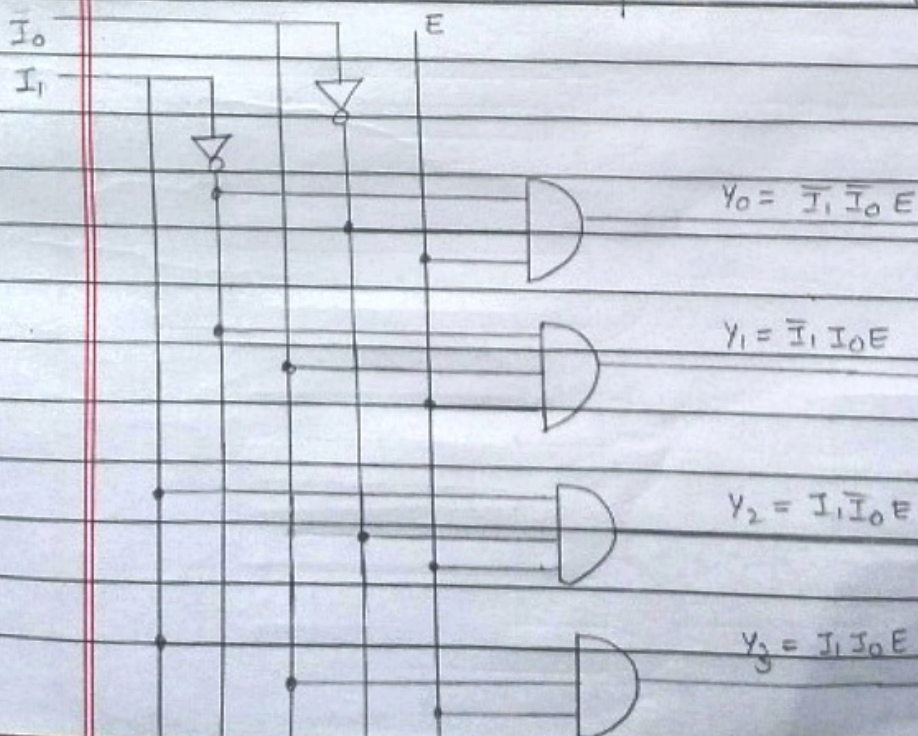
If we interchange i/p and selected lines (S.L) of demultiplexer it becomes decoder. $E \rightarrow$ enable line

	i/p	Total o/p's = 2
1/2 : o/p		
2 : 4		3:8 decoder
Decoder		4:16 "
		5:32 "

Active High decoder

Every combinational circuit has enable line. Decoder works only when $E=1$.

E	I_1	I_0	Y_3	Y_2	Y_1	Y_0	
0	x	x	invalid.				$Y_0 = \bar{I}_1 \bar{I}_0 E$
1	0	0	0	0	0	1	$Y_1 = \bar{I}_1 I_0 E$
1	0	1	0	0	1	0	$Y_2 = I_1 \bar{I}_0 E$
1	1	0	0	1	0	0	$Y_3 = I_1 I_0 E$
1	1	1	1	0	0	0	



Implement $Y(A,B,C) = \sum m(1,2,6,7)$ or $Y(A,B,C) = \prod M(0,3,4,5)$
using

(a) 8:1 MUX

(c) Active high decoder

(e) DMUX.

(b) 4:1 MUX

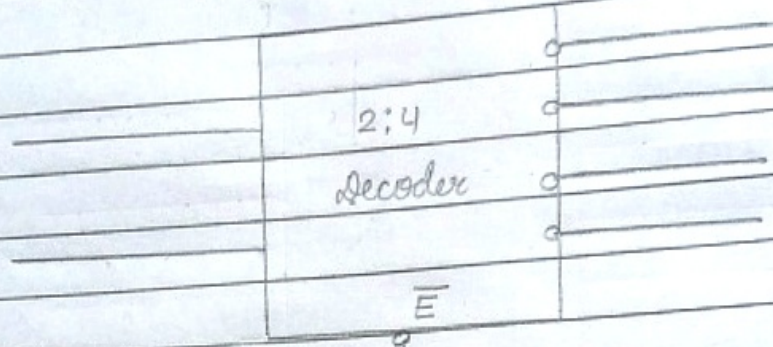
(d) Active low decoder

MUX $\rightarrow$ AND-OR

DMUX }
Decoder } AND.

Active low decoder

it'll work only if $\bar{E} = 0$.



$\bar{E}$	I_1	I_0	Y_3	Y_2	Y_1	Y_0
1	x	x	Invalid			
0	0	0	1	1	1	0
0	0	1	1	1	0	1
0	1	0	1	0	1	1
0	1	1	0	1	1	1

Ques. Implement following functions using

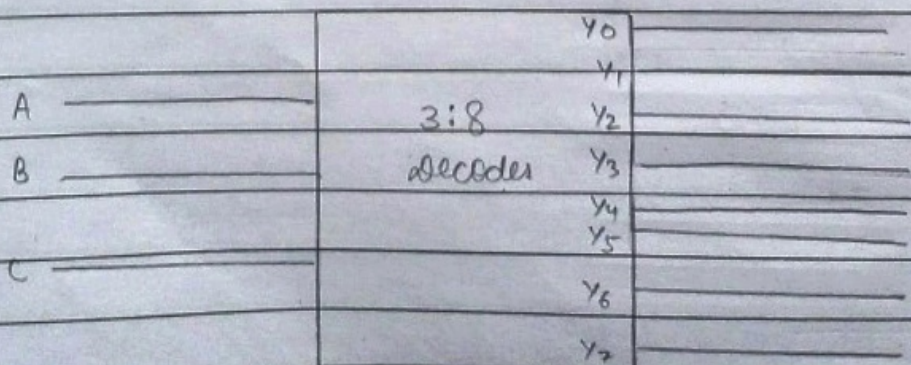
(a) Active high decoder (by default)

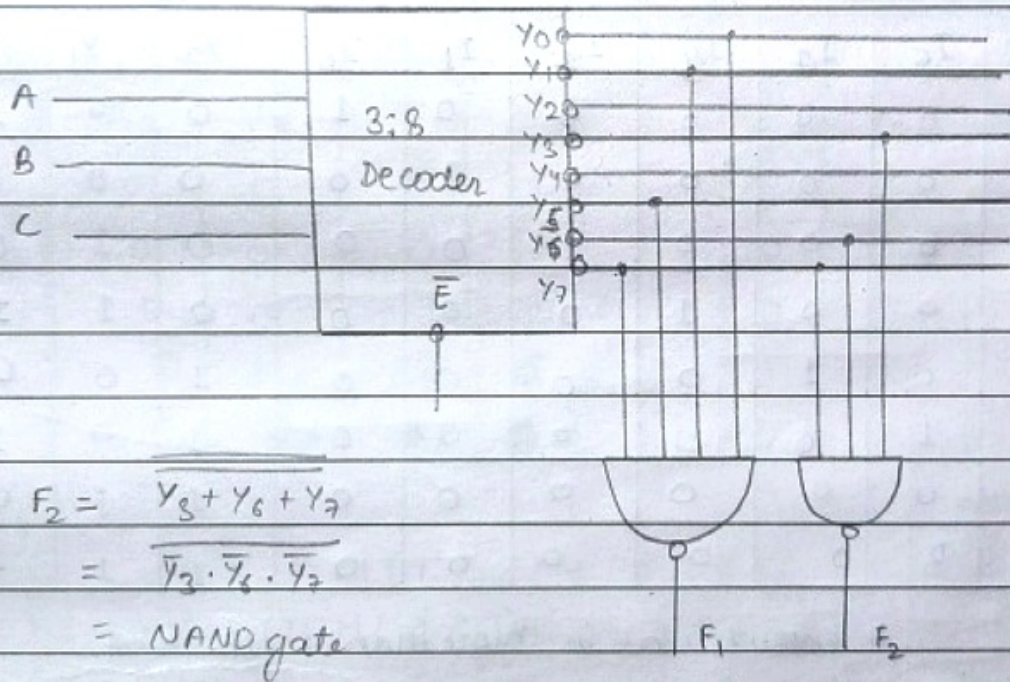
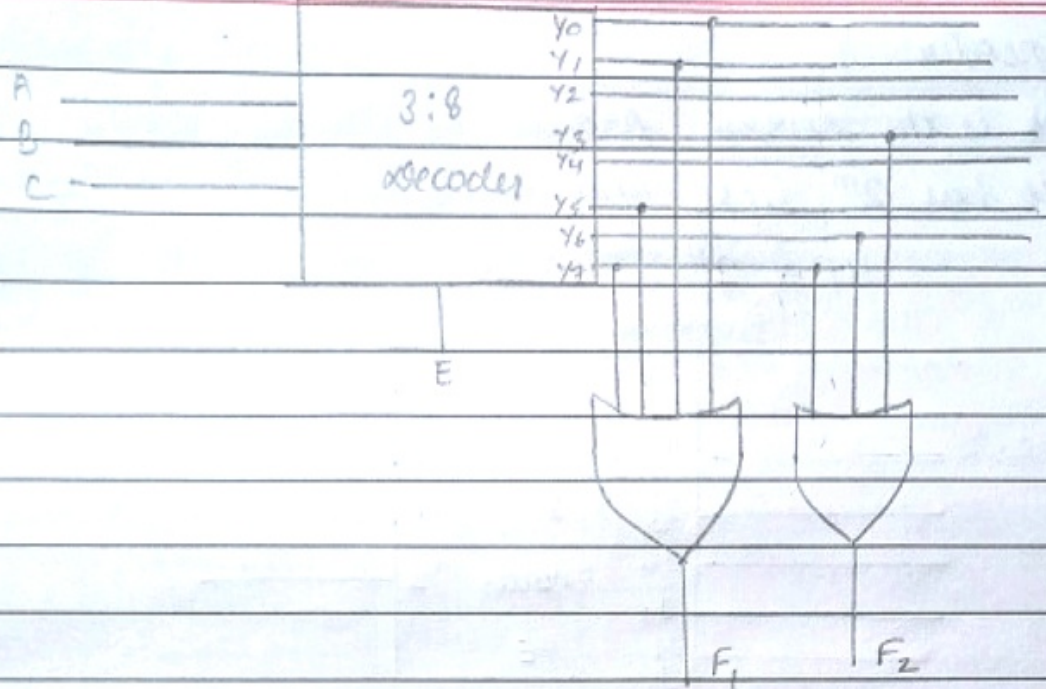
(b) Active low decoder

$$F_1(A, B, C) = \sum m(0, 1, 5, 7)$$

$$O/P = 2^{i/P}$$

$$F_2(A, B, C) = \pi M(0, 1, 2, 4, 5)$$





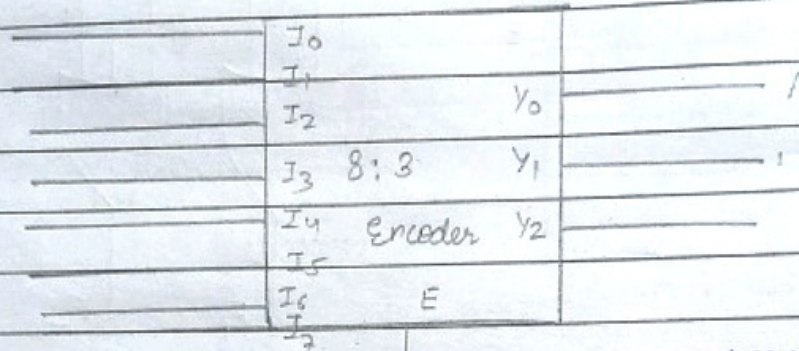
Decoder converts binary code to other codes (octal or hexadecimal)

Encoder

It is the reverse process of decoder.

It has 2^m i/p's and m o/p's

$$\text{i/p} = 2^{\text{o/p}}$$



								(msb)		
I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

It doesn't work in more than 1 i/p. high.

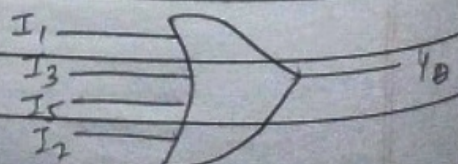
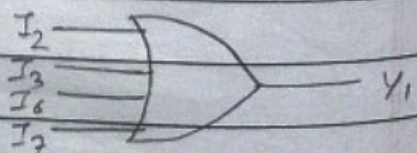
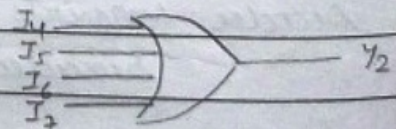
Limitation: If all i/p's are 0, no output is attained.

अगर 1 है, उसका binary equivalent in Y_2, Y_1, Y_0 .

$$Y_2 = I_4 + I_5 + I_6 + I_7$$

$$Y_1 = I_2 + I_3 + I_6 + I_7$$

$$Y_0 = I_1 + I_3 + I_5 + I_7$$



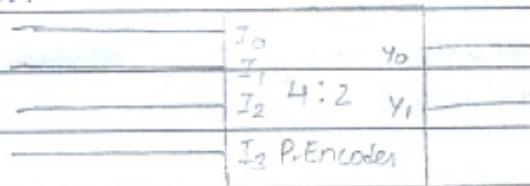
NOTE: Encoder converts other code (octal / hexadecimal) to binary.

Priority Encoder

In normal encoder,

- i. If all i/p.s are low,
- ii. If more than 1 i/p.s are high we don't get any valid output.

This problem / limitation is avoided by priority encoder.



don't care

← Priority order

I_3	I_2	I_1	I_0	Y_1	Y_0	Valid (V)
0	0	0	0	X	X	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

I_1
1 1
1 0

I_2
1 0 0
1 0 1
1 1 1

4
5
6
7

$$Y_1 = \sum m(1, 2, 3) + \sum d(0)$$

$$Y_0 = \sum m(0, 2, 1, 4, 5, 6, 7) + \sum d(0)$$

$$V = M0 = (I_3 + I_2 + I_1 + I_0)$$

X will have either 0 or 1.

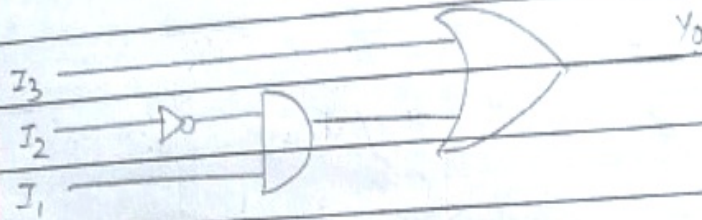
$I_2 I_1$	$I_3 I_0$	00	01	11	10
00	X	0	0	0	0
01					
11					
10					

$I_2 I_1$	$I_3 I_0$	00	01	11	10
00	X	0	0	0	0
01	0	0	0	0	0
11					
10					

$$Y_1 = I_2 + I_3$$

$$Y_0 = (I_3 + \bar{I}_2)(I_1 + I_3)$$

$$Y_0 = I_3 + I_1 \bar{I}_2$$



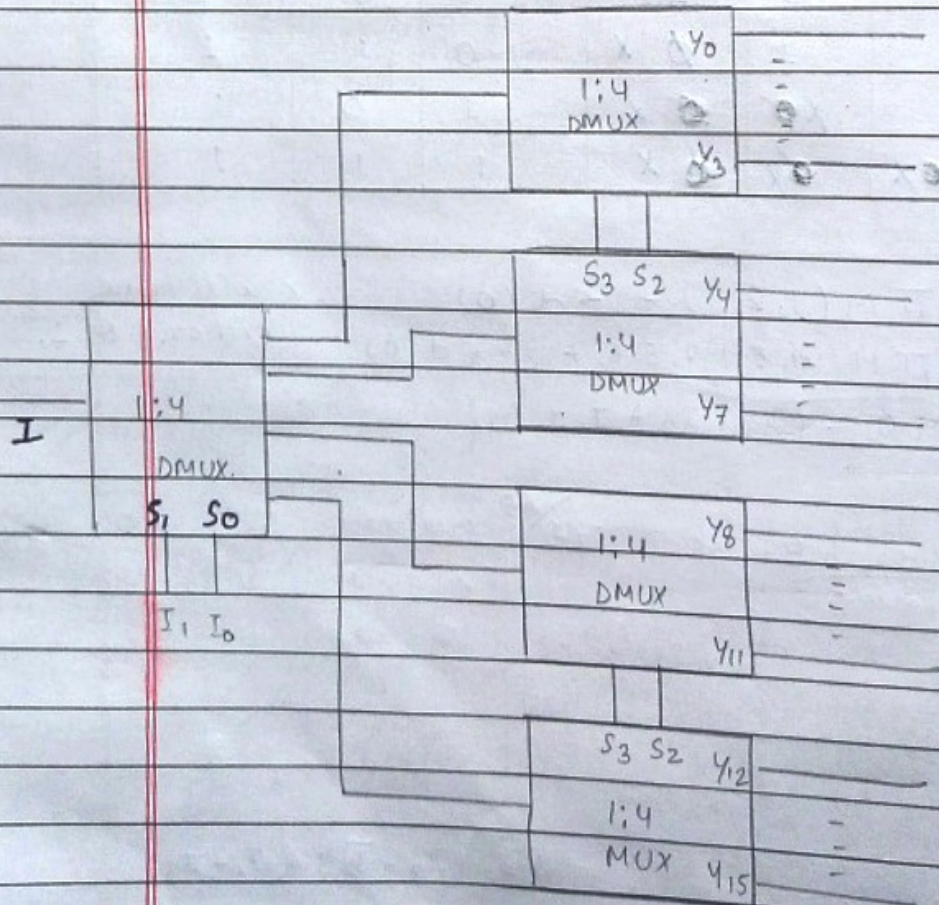
Ques. (a) Implement 1:16 DMUX using 1:4 DMUX.

(b) Implement 4:16 decoder using 2:4 Decoder.

(c) Implement 3:8:3 encoder using 2:2 and suitable encoder.

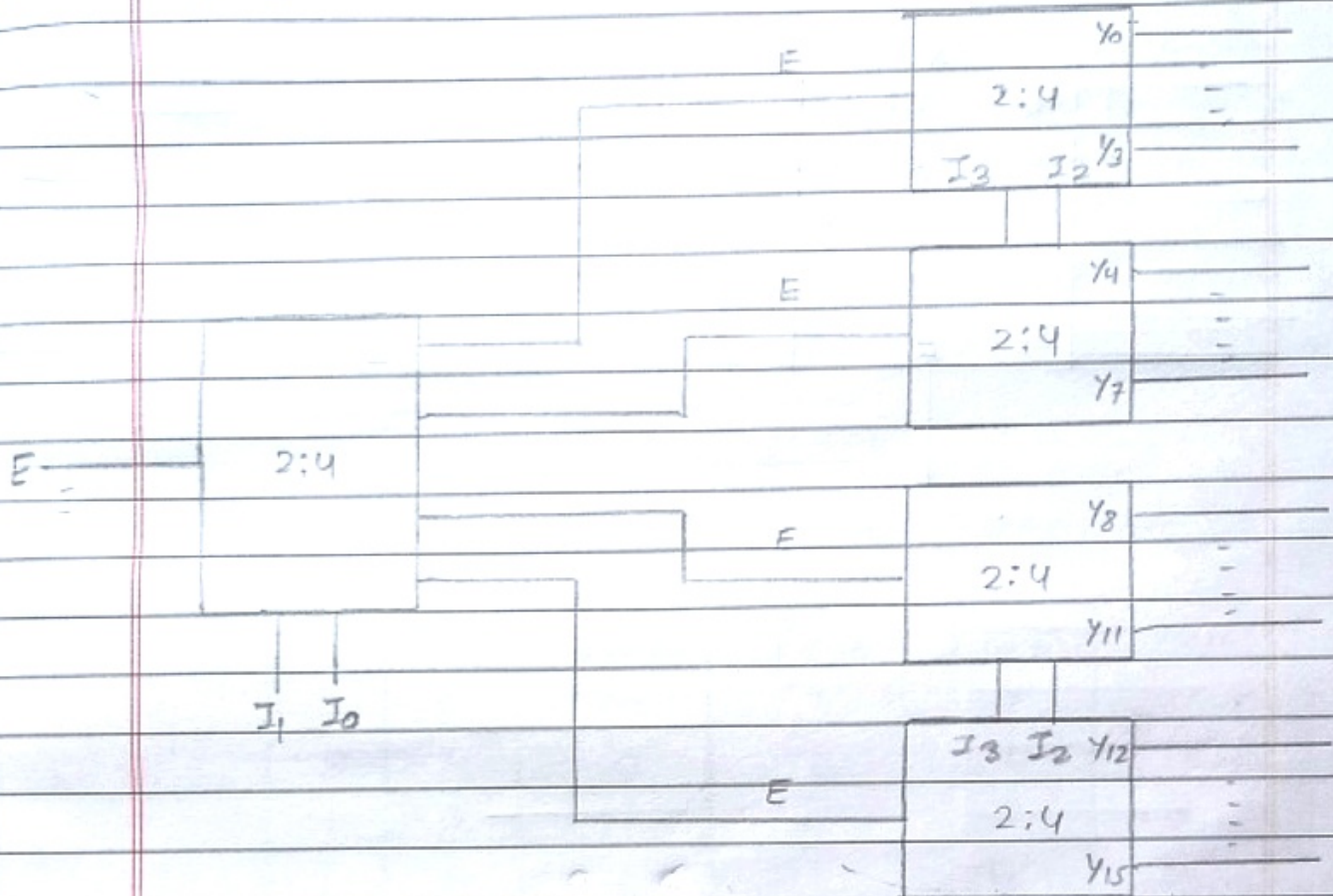
(a) 1:16. 1:4.

$$\frac{16}{4} = \frac{4}{1} = 4$$

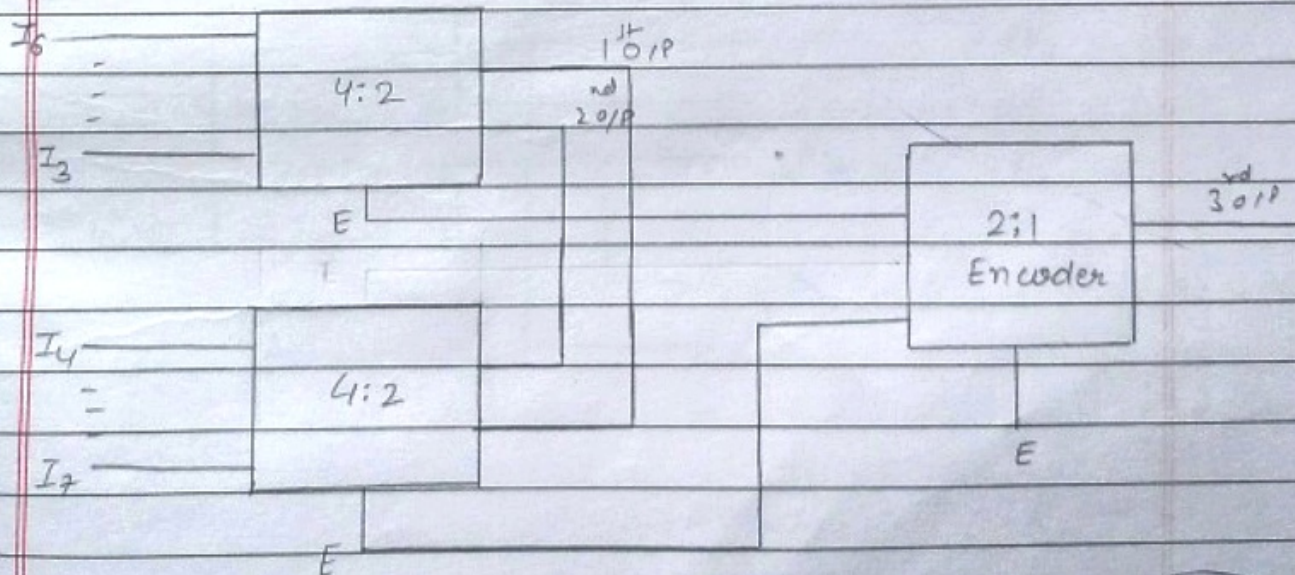


4+3=5 decoder

(b) $\frac{16}{4} = \frac{4}{4} = 1$



(c) $\frac{8}{4} = \frac{2}{2} = 1$ (2) 4:2 + (1) 2:1



Significance of Encoder & Decoder

Tx			Rx		
0	I ₀			Y ₀	
0					
0	8:3	Y ₂	3:8		
1	Encoder	Y ₁	Decoder		
0					
0		Y ₀			
0					
6	I ₂	E	E	Y ₇	